NAMIBIA UNIVERSITY OF SCIENCE AND TECHNOLOGY
Faculty of Computing and Informatics
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COURSE OUTLINE DESIGN THINKING (DST511S)

STATEMENT ABOUT ACADEMIC HONESTY AND INTEGRITY

All staff and students of the Namibia University of Science and Technology (NUST), upon signing their employment contracts and registration forms, commit themselves to abide by the policies and rules of the institution. The core activity of NUST is learning and in this respect academic honesty and integrity is very important to ensure that learning is valid, reliable and credible.

NUST therefore does not condone any form of academic dishonesty, including plagiarism and cheating on tests and assessments, amongst other such practices. NUST requires students to always do their own assignments and to produce their own academic work, unless given a group assignment.

Academic Dishonesty includes, but is not limited to:

- Using the ideas, words, works or inventions of someone else as if it is your own work.
- Using the direct words of someone else without quotation marks, even if it is referenced.
- Copying from writings (books, articles, webpages, other students' assignments, etc.), published or unpublished, without referencing.
- Syndication of a piece of work, all or part of an assignment, by a group of students, unless the assignment was a legitimate group assignment.
- The borrowing and use of another person's assignment, with or without their knowledge or permission.
- Infringing copyright, including documents copied or cut and pasted from the internet.
- Asking someone else to prepare an assignment for you or to write or sit an assessment for you, whether this is against payment or not.
- Re-submitting work done already for another course or programme as new work, so-called self-plagiarism.
- Bringing notes into an examination or test venue, regardless of whether the notes were used to copy or not.
- Receiving any outside assistance in any form or shape during an examination or test.

All forms of academic dishonesty are viewed as misconduct under NUST Student Rules and Regulations. Students who make themselves guilty of academic dishonesty will be brought before a Disciplinary Committee and may be suspended from studying for a certain time or may be expelled. All students who are found guilty of academic dishonesty shall have an appropriate endorsement on their academic record, which will never be erased.

COURSE INFORMATION

COURSE CODE AND TITLE: DST511S DESIGN THINKING

DEPARTMENT: SOFTWARE ENGINEERING

PROGRAMME: Bachelor of Informatics, Bachelor of Computer Science

CONTACT HOURS: 80 hours divided as follows:

30 hours of face-to-face between students and lecturers;
30 hours of directed Self-learning and self-directed learning;
20 hours for assessment (group discussions, preparing presentations)

NQF LEVEL AND CREDIT: Level 5, 8 Credits

COURSE DESCRIPTION:

This course aims to equip students with Design Thinking concepts and tools synthesising human-centred techniques to solve problems in a more creative and innovative approach. The course concentrates on evaluating alternative methods to solve problems that might not be instantly evident. Students will be able to integrate the iterative design thinking process of empathising with users of the product or service, defining ideas, producing a prototype and testing.

PRE-REQUISITES: None

COURSE EQUIVALENCIES:

None

COURSE DELIVERY METHODS:

The course will be facilitated through blended learning lectures, classroom presentations, theory classes' online and practical sessions face-to-face on campus, case studies, discussions, assignments and exercises.

The following communication tools will be used in this course:

Moodle, Email, Zoom, Microsoft Teams & WhatsApp

Course Format:

Classroom presentations, theory and practical demonstrations, self-guided learning, student presentations and exercises and hands-on application of the design thinking process.

EFFECTIVE DATE:

26 February 2025

LECTURER INFORMATION

Ms Rosetha Kays (Course Coordinator) Phone: 061 207 2405 Office: Poly-Heights, 4th Floor, Room 4 E-mail: rkays@nust.na	Dr Gabriel Nhinda (Practical Lecturer) Phone: 061 207 2724 Office: HTTPS, 2nd Floor E-mail: gnhinda@nust.na
Ms Iitumba, Ndinelao (Practical Lecturer) E-mail: niitumba@nust.na	Ms Laurinda Ndjao E-mail: ndjaolaurindapandula@gmail.com
Ms Auala, Selma E-mail: seauala@nust.na	Ms Minikuee Kasaona E-mail: minikuee@gmail.com
Mr Andrew Tjirare E-mail: atjirare@nust.na	Mr Sakaria Nghivafe E-mail: sakarianghivafe@gmail.com

Consultation hours: All lecturers are available for consultation with students during office hours or on appointment.

STUDENT READINESS

Technology & Equipment Readiness:

Students should be able to use the different ICT devices to access Moodle (e-learning), MS Teams and other communicate online communication platforms.

Student Commitments and Contact Times:

Students should attend online classes and participate in all online discussions and course activities as will be indicated on Moodle.

Course Resources:

All resources necessary for the course will be available for the students. All lecture notes and course materials will be available on on Moodle (E-learning).

Prescribed Reading:

Vianna, M., Vianna, Y., Adler, I., Lucena, B., & Russo, B. (2012). *Design thinking: Business innovation*. MJV Tecnologia ltda, Rio de Janeiro.

Brown, T. & Katz, B. (2019*). Change by design: how design thinking transforms organizations and inspires innovation*. New York: HarperBusiness, an imprint of HarperCollinsPublishers.

Recommended Reading:

Eppinger, Steven D., & Karl T. Ulrich. (1995). "Product design and development."

Liedtka, J., Ogilvie, T., & Brozenske, R. (2014). *The designing for growth field book: A step-by-step project guide.* Columbia Uni-versity Press.

Liedtka, J., & Ogilvie, T. (2011). *Designing for growth: A design thinking tool kit for managers*. Columbia University Press.

Liedtka, J., Salzman, R., & Azer, D. (2017). *Design thinking for the greater good: Innovation in the social sector*. Columbia University Press.

Rowe, P. G. (1991). *Design thinking*. MIT press.

Nieman, G. & Pretorius, M. (2004). *Managing Growth. A guide for entrepreneurs*. Juta Academic: Cape Town.

STUDENT LEARNING

Learning Outcomes:

By the end of this course of study, you should be able to:

- Analyse the problem from a customer's perspective (human-centred problem solving) and understand that innovation does not always involves a product;
- Brainstorm effectively innovative ideas;
- Create and evaluate prototype designs;
- Develop skills to work with diverse inclusive\exclusive teams;
- Plan and manage innovation projects and processes effectively (and measure their success against; innovation metrics such as throughput, success rate and value;
- Apply design thinking to real-life problems within the Namibian context.

COURSE CONTENT

	Topic	**Assessments/Practical**
Week 1 10 - 14 Feb	Introduction • Course Outline Introduction • Introduction to Design Thinking	**Online Activity:** introduce yourself via Moodle (e-learning.nust.na) Mandatory reading
Week 2 17-21 Feb	Understanding Design Thinking • Design Thinking Methodology: **Empathize Stage** • Identifying Customer Needs • Methods within the empathise stage	Practical (Face-to-Face): Understand the problem and the affected humans. Group formation Release Group project
Week 3 24-28 Feb	Design Thinking Methodology: **Define Stage** • Personas	Practical (Face-to-Face): Synthesize the findings from the Empathize stage.
Week 4 3-7 Mar	Design Thinking Methodology: **Ideate Stage**	Practical (Face-to-Face): Generating a wide range of ideas and potential solutions to address the problem. • Submission: Idea and Problem Statement presentation.
Week 5 10-14 Mar	Design Thinking Methodology: **Prototype Stage** • Paper • Digital • Tools	Practical (Face-to-Face): Mind mapping and Problem voting Practical Activity ***Quiz 1***
Week 6 17-21 Mar	Design Thinking Methodology: **Test/ Evaluation** • Methods • Online student Orientation to Moodle and fixing modalities	Practical (Face-to-Face): Prototype Design

21 March Independence Day		
Week 7 24-28 Mar	Design Thinking Methodology: **Iterate**	Practical (Face-to-Face): Prototyping
Mid Semester Break *Week 8* 31 Mar – 04 Apr		
Week 9 7-11 Apr	• Emotional Intelligence • Ethics in Design Thinking • Population • Power Dynamics	Practical (face-to-face): Prototype testing/evaluate the prototypes you've created and gather feedback from users. ***Quiz 2***
Week 10 14-17 Apr 18 Apr Public Holiday (Good Friday)	• Design Processes • User Centered Design • Human Centered Design • Participatory Design • Co-Creation • Collaborative design	Practical (Face-to-Face): design processes case studies
Week 11 21–25 Apr 21 Apr Public Holiday (Easter Monday)	• Prototype Testing - Refining • Improving new ideas and processes	**Due date: Final Project Poster and Project Exhibition Video:** **25 April 2025**
Week 12 28 Apr – 2 May 1 May Thursday (Workers Day)	• User Experience • Innovation Frameworks	Practical (Face-to-Face): Remote Usability labs for prototypes Pitch
Week 13 5-9 May Monday 5 May (Public Holiday)	• Revision	**Supplementary Quiz**
Week 14 12–16 May	• Conclusions and Debriefing	
Week 15 19 May - 6 June	• Student engagement (no practical) • Students Verify Semester Marks	**Project Exhibitions – NUST Open Area**
First Semester Lectures End for CASS Courses 06 June		

IMPORTANT DATES:

NOTE: The following dates are subject to change based on the needs of the students at the lecturer's prerogative. Students will be notified ahead of time of any changes.

Assessment	Date	Venue
Quiz 1	14 March	Online (Moodle)
Quiz 2	11 April	Online (Moodle)
Supp Quiz 3	9 May	Online (Moodle)
Idea and problem statement presentation	03 March – 07 March	Face-to-Face Online submission
Final Poster	25 April	Practical Venue + Online Submission
Pitch (high fidelity prototype)	2 May	Face-to-face
Project Exhibition video	25 May	Online Submission

ASSESSMENT AND EVALUATION:

Assessment	Weights	Venue
Quiz 1	10%	Online (Moodle)
Quiz 2	10%	Online (Moodle)
Supplementary Quiz 3	Replace any missed quiz	Online (Moodle)
Idea and problem statement presentation	20%	Face-to-Face Online submission
Pitch (high fidelity prototype)	20%	Face-to-face
Final Poster	20%	Practical Venue + Online Submission
Project Exhibition video	20%	Practical Venue + Online Submission
Course total	100%	

Minimum pass requirement for this course: A minimum Final Mark of 50% is required to pass the course.

COURSE POLICIES

General Academic Policies:

It is the student's responsibility to be familiar with and adhere to NUST's Policies. These Policies can be found in NUST Prospectus or online at www.nust.na/prospectus.

Supplementary Policies:

Two copies of assignments should be submitted to the class lecturer. One should be a printed hard copy. The soft copy must be saved in the drive that will be indicted on the assignment question paper. All Late Assignments will not be accepted after the due date and time and thus marked as 0%.

DATE REVISED: 26 February 2025

FAILURE TO PAY FEES:

A student who fails to pay his/her fees may not be allowed to write the examination and if allowed, the results will be withheld until all outstanding fees are paid in full.

IMPORTANT STUDENT SERVICES AT NUST

There are a variety of services which you can use at the NUST. These services are to your advantage – Use them!!! They include the following:

- Student Counseling and Career Development – Department: Students Services
- Writing Centre and student academic problems –Teaching and Learning Unit (CTL)
- Campus Health and Wellness Centre (CHWC) - Student Services'/ NUST Clinic

AUTHORISATION:

This course is authorised for use by:

SM _____ 26 February 2025

Head of Department Date

ACKNOWLEDGEMENT BY STUDENT

(To be completed by all students on the course, detached from the course outline and kept on record in the department)

I, (insert name), (Student number), hereby acknowledge that I have received this course outline for (insert course title and code), and that I have familiarised myself with its content, in particular the statement about academic honesty and integrity. I agree to abide by the Policies and arrangements spelt out in this course outline.

_____ _____

Signature of Student